

Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

Question 1: Mapping Data onto Aesthetics

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

Questions

1. Write an R program to create a scatter plot with Sales on the x-axis and Profit on the y-axis.

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Product = c("P1","P2","P3","P4","P5","P6"),
```

```
  Sales = c(120,90,150,80,135,100),
```

```
Profit = c(18,12,25,8,22,16),
```

```
Category =
```

```
c("Electronics","Clothing","Electronics","Furniture","Furniture","Clothing"))
```

```
ggplot(data, aes(x = Sales, y = Profit,
```

```
color = Category,
```

```
size = Sales)) +
```

```
geom_point() +
```

```
labs(title = "Sales vs Profit",
```

```
x = "Sales (₹ Lakhs)",
```

```
y = "Profit (%)")
```

2. Map Category using an appropriate aesthetic and Sales using another suitable aesthetic.

Aesthetic Mapping

- X-axis → Sales
- Y-axis → Profit
- Color → Category
- Size → Sales

3. Interpret the relationship between Sales and Profit.

Interpretation

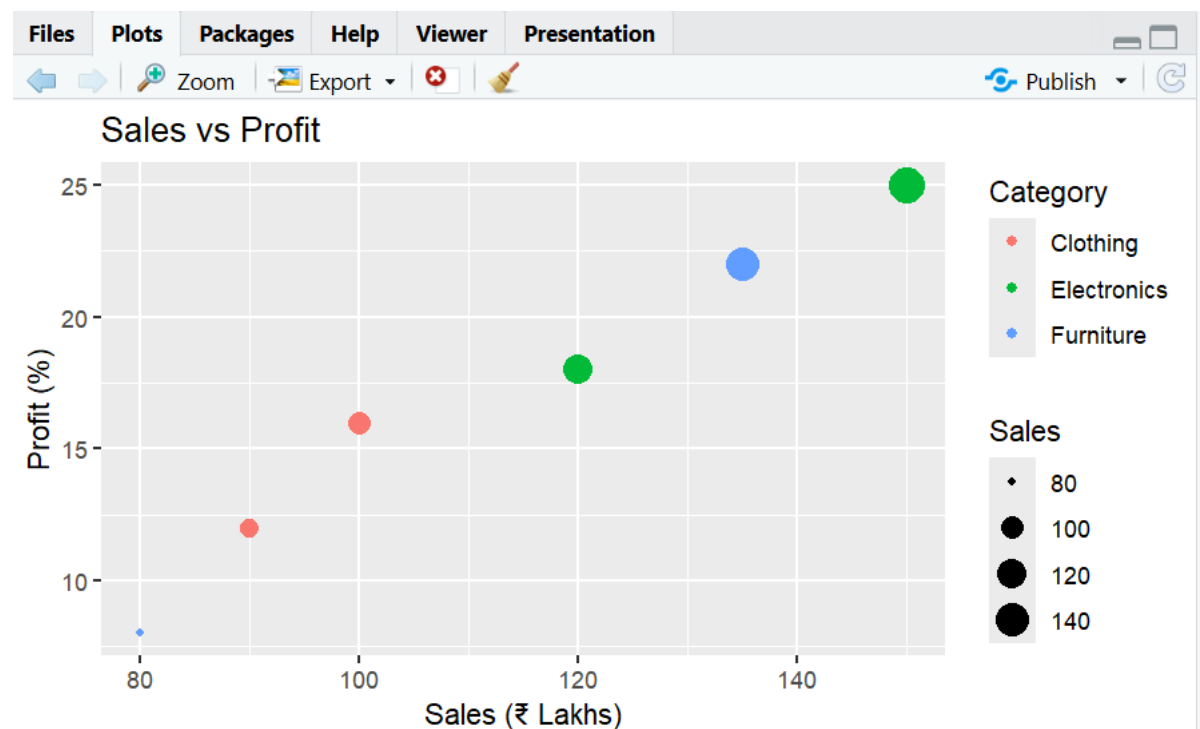
- Sales and Profit have a **positive relationship**.
- Products with higher sales generally have higher profit.
- Electronics products show the highest sales and profit.

4. Identify the type of scale used for each variable.

Type of Scale

- Sales → Continuous scale
- Profit → Continuous scale
- Category → Categorical (Discrete) scale

5. Suggest one alternative visualization that better represents the same data.



Question 2: Coordinate Systems (Cartesian and Polar)

Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11
W	16
NW	14

Questions

1. Write an R program to visualize the dataset using a Cartesian coordinate system.

```
library(ggplot2)

direction <- data.frame(

  Direction = c("N","NE","E","SE","S","SW","W","NW"),

  Frequency = c(18,12,20,15,22,11,16,14))

ggplot(direction, aes(Direction, Frequency)) +

  geom_bar(stat="identity", fill="steelblue")
```

2. Write another R program to visualize the same data using a Polar coordinate system.

```
ggplot(direction, aes(x="", y=Frequency, fill=Direction)) +
```

```
geom_bar(stat="identity", width=1) +  
coord_polar("y")
```

3. Compare the effectiveness of both visualizations.

Comparison

Cartesian

- Easy to compare frequencies.
- More accurate.

Polar

- Attractive appearance.
- Shows circular directions naturally.

4. Explain which coordinate system is more appropriate for directional data.

More Appropriate Coordinate System

Polar Coordinate System

5. Justify your answer based on data interpretation.

Justification

Because the data represents **compass directions**, polar coordinates naturally represent directional information.

Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72

Student	Marks
D	81
E	67
F	90

Questions

1. Write an R program to create a bar chart using a sequential color scale based on Marks.

```
library(ggplot2)

student <- data.frame(

  Student=c("A","B","C","D","E","F"),

  Marks=c(45,58,72,81,67,90))

ggplot(student, aes(Student, Marks, fill=Marks)) +

  geom_bar(stat="identity") +

  scale_fill_gradient(low="lightblue", high="dark blue")
```

2. Identify the students with the highest and lowest marks from the visualization.

Highest and Lowest Marks

- Highest → **Student F (90)**
- Lowest → **Student A (45)**

3. Explain why a sequential color scale is appropriate.

Because marks are **continuous numerical values**, darker colors indicate higher marks while lighter colors indicate lower marks.

4. Suggest a color palette suitable for color-blind users.

Color-blind Friendly Palette

Viridis palette

`scale_fill_viridis_c()`

5. Explain the difference between sequential and categorical color scales.

Sequential Scale	Categorical Scale
Used for numerical data	Used for categories
Colors change gradually	Different colors represent different groups

Question 4: Highlighting with Color

Dataset

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

Questions

1. Write an R program to create a bar chart highlighting the region with the highest revenue.

```
library(ggplot2)

region <- data.frame(

  Region=c("North","South","East","West","Central"),

  Revenue=c(820,760,790,1010,770))

region$Highlight <-

  ifelse(region$Revenue==max(region$Revenue),"Highest","Other")

ggplot(region,

  aes(Region, Revenue, fill=Highlight))+

  geom_bar(stat="identity")+

  scale_fill_manual(values=c("red","gray"))
```

2. Explain how highlighting improves data interpretation.

How Highlighting Helps

- Draws attention to important values.
- Makes comparison easier.
- Improves readability.

3. Identify the highest-performing region.

Highest Performing Region

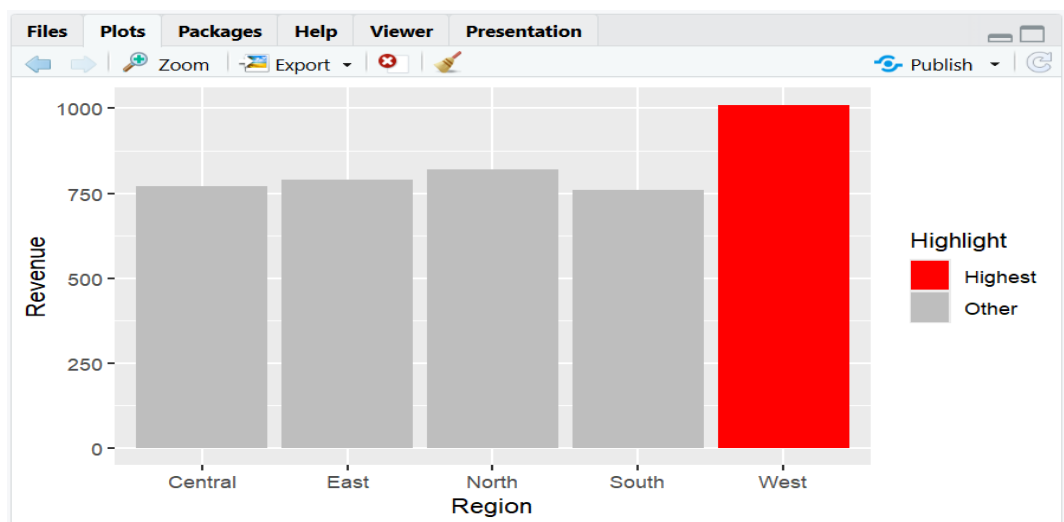
West (₹1010 Crores)

4. Discuss how excessive use of colors can affect interpretation.

Effect of Excessive Colors

- Confuses viewers.
- Reduces clarity.
- Makes important information difficult to identify.

5. Suggest one alternative highlighting technique other than color.



Question 5: Integrated Visualization

Dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6		430	South
C		180		
D	9.4		1200	North
E	6.8	90	520	East
	4.9	240	680	West
		150		

Questions

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

```
library(ggplot2)
```

```
city <- data.frame(
```

```
  City=c("A","B","C","D","E"),
```

```
  Population=c(8.2,5.6,9.4,6.8,4.9),
```

```
  AQI=c(110,180,90,240,150),
```

```
  Rainfall=c(850,430,1200,520,680),
```

```
  Region=c("North","South","North","East","West"))
```

```
ggplot(city,
```

```
  aes(x=Population,
```

```
    y=AQI,
```

```
    size=Rainfall,
```

```
color=Region))+  
geom_point(alpha=0.7)+  
labs(title="Population vs AQI")
```

2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

Suitable Aesthetic Mapping

- X-axis → Population
- Y-axis → AQI
- Bubble Size → Rainfall
- Color → Region

3. Interpret the relationship between Population and AQI.

Interpretation

- Population does **not always increase AQI**.
- City D has moderate population but the highest AQI.
- Other factors besides population affect air quality.

4. Identify the continuous and categorical variables.

Continuous and Categorical Variables

Continuous Variables

- Population
- AQI

- Rainfall

Categorical Variables

- City
- Region

5. Recommend one improvement to enhance the visualization's readability.

Improvement

- Add **data labels** (city names) to each bubble for easier identification.